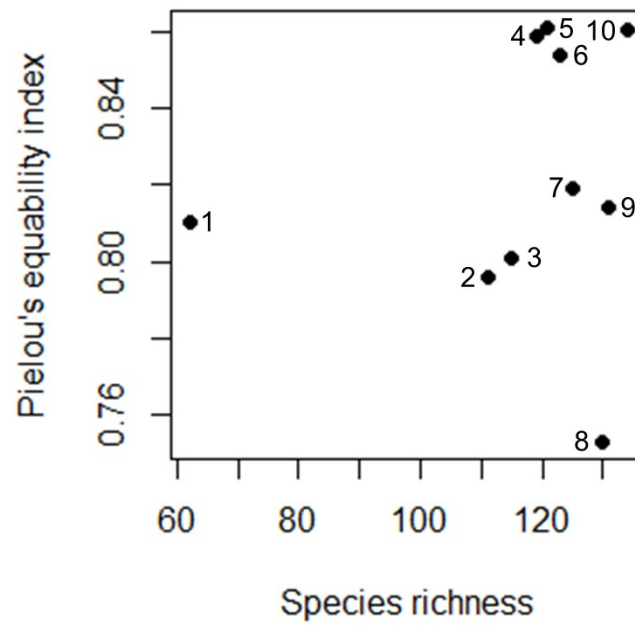


Online Resource 1. Sampling sites in the city of Curitiba, Southern Brazil.

Codes	Site name	Latitude	Longitude
1	Passeio Público	-25.425419	-49.267331
2	Parque Tingui	-25.396317	-49.305548
3	Campus Jardim Botânico – UFPR	-25.446535	-49.237177
4	Parque Atuba	-25.379862	-49.207268
5	Quartel 5º GAC AP	-25.507646	-49.239813
6	Parque Regional de Manutenção/5	-25.398833	-49.224716
7	Quartel General 5ª DE	-25.526246	-49.308808
8	Parque Tanguá	-25.37875	-49.28387
9	Cemitério Parque Iguaçu	-25.419457	-49.31182
10	Aeroporto Internacional Afonso Pena	-25.525735	-49.171811

Online Resource 2. Dispersion diagram between species richness and Pielou equability index (J') (Pielou, 1966) for each sample area.



Online Resource 3. Values of the coefficient of determination (R^2) for each of the bee responses and the landscape predictors that showed to be important in the model selection analysis, for all the studied scales.

Responses	Landscape Predictors	Scale	R^2
Bee Diversity	Proportion of vegetation cover	0.75	0.250
		1	0.161
		1.5	0.103
		2	0.044
	Isolation	0.75	0.265
		1	0.089
		1.5	0.061
		2	0.003
	Landscape diversity	0.75	0.274
		1	0.184
		1.5	0.131
		2	0.079
Bee Richness	Proportion of grass cover	0.75	0.383
		1	0.337
		1.5	0.311
		2	0.228
	Proportion of vegetation cover	0.75	0.668
		1	0.569
		1.5	0.436
		2	0.359
	Isolation	0.75	0.532
		1	0.365
		1.5	0.335
		2	0.212
	Landscape diversity	0.75	0.821
		1	0.711
		1.5	0.588
		2	0.488
Solitary Richness	Proportion of grass cover	0.75	0.352
		1	0.314
		1.5	0.288
		2	0.329
	Proportion of vegetation cover	0.75	0.652
		1	0.571
		1.5	0.464
		2	0.376
	Isolation	0.75	0.478
		1	0.418
		1.5	0.343
		2	0.230
	Landscape diversity	0.75	0.771
		1	0.691
		1.5	0.604
		2	0.505
	Proportion of grass cover	0.75	0.384
		1	0.361
		1.5	0.360
		2	0.399

Responses	Landscape Predictors	Scale	R ²
Eusocial Richness	Proportion of vegetation cover	0.75	0.238
		1	0.197
		1.5	0.147
		2	0.173
	Isolation	0.75	0.321
		1	0.053
		1.5	0.236
		2	0.135
	Landscape diversity	0.75	0.509
		1	0.394
		1.5	0.242
		2	0.221
Primitively Eusocial Richness	Proportion of grass cover	0.75	0.032
		1	0.017
		1.5	0.000
		2	0.013
	Proportion of vegetation cover	0.75	0.443
		1	0.346
		1.5	0.268
		2	0.266
	Isolation	0.75	0.339
		1	0.053
		1.5	0.131
		2	0.122
Highly Eusocial Richness	Landscape diversity	0.75	0.616
		1	0.463
		1.5	0.303
		2	0.304
	Proportion of grass cover	0.75	0.312
		1	0.275
		1.5	0.175
		2	0.232
	Proportion of vegetation cover	0.75	0.001
		1	0.000
		1.5	0.000
		2	0.001
Highly Eusocial Richness	Isolation	0.75	0.036
		1	0.007
		1.5	0.103
		2	0.024
	Landscape diversity	0.75	0.032
		1	0.028
		1.5	0.012
		2	0.006
	Proportion of grass cover	0.75	0.141
		1	0.166
		1.5	0.212
		2	0.149

Responses	Landscape Predictors	Scale	R ²
Ground nest Richness	Proportion of vegetation cover	0.75	0.754
		1	0.678
		1.5	0.558
		2	0.484
	Isolation	0.75	0.529
		1	0.416
		1.5	0.403
		2	0.334
	Landscape diversity	0.75	0.774
		1	0.711
		1.5	0.643
		2	0.591
Above Ground Nests Richness	Proportion of vegetation cover	0.75	0.110
		1	0.074
		1.5	0.049
		2	0.030
	Isolation	0.75	0.172
		1	0.100
		1.5	0.091
		2	0.004
	Landscape diversity	0.75	0.426
		1	0.312
		1.5	0.184
		2	0.090
Specialist Richness	Proportion of grass cover	0.75	0.002
		1	0.002
		1.5	0.001
		2	0.000
	Proportion of vegetation cover	0.75	0.793
		1	0.733
		1.5	0.619
		2	0.562
	Isolation	0.75	0.376
		1	0.391
		1.5	0.361
		2	0.372
	Landscape diversity	0.75	0.707
		1	0.684
		1.5	0.677
		2	0.665
	Proportion of grass cover	0.75	0.519
		1	0.486
		1.5	0.470
		2	0.563

Responses	Landscape Predictors	Scale	R ²
Generalist Richness	Proportion of vegetation cover	0.75	0.519
		1	0.433
		1.5	0.338
		2	0.267
	Isolation	0.75	0.542
		1	0.343
		1.5	0.336
		2	0.162
	Landscape diversity	0.75	0.808
		1	0.675
		1.5	0.515
		2	0.391
Cleptoparasitic Richness	Proportion of grass cover	0.75	0.234
		1	0.211
		1.5	0.190
		2	0.218
	Proportion of vegetation cover	0.75	0.469
		1	0.314
		1.5	0.151
		2	0.096
	Isolation	0.75	0.417
		1	0.154
		1.5	0.093
		2	0.025
	Landscape diversity	0.75	0.457
		1	0.316
		1.5	0.225
		2	0.156
	Proportion of grass cover	0.75	0.239
		1	0.169
		1.5	0.134
		2	0.118

Online Resource 4. Bee species and sampling sites in the city of Curitiba, Southern Brazil, site codes according Online Resource 1.

Bee species	1	2	3	4	5	6	7	8	9	10	Total
Andreninae											
Calliopsini											
<i>Acamptopoeum prini</i> (Holmberg, 1884)	0	0	0	0	1	2	0	0	0	0	3
<i>Callonychium</i> (<i>Callonychium</i>) <i>petuniae</i> Cure & Wittmann, 1990	0	0	0	0	0	0	1	0	0	0	1
Protandrenini											
<i>Anthrenoides admirabilis</i> Urban, 2005	0	0	0	0	0	0	0	1	0	0	1
<i>Anthrenoides alvarengai</i> Urban, 2007	0	0	0	1	0	0	0	0	1	0	2
<i>Anthrenoides antonii</i> Urban, 2005	0	0	0	0	0	0	4	0	0	2	6
<i>Anthrenoides araucariae</i> Urban, 2005	0	0	0	0	0	0	0	9	0	0	9
<i>Anthrenoides corrugatus</i> Urban, 2005	0	0	0	0	0	0	0	1	0	0	1
<i>Anthrenoides larocai</i> Urban, 2005	0	0	0	0	0	0	0	0	0	1	1
<i>Anthrenoides meridionalis</i> (Schrottky, 1906)	12	3	4	7	6	11	1	5	11	3	63
<i>Anthrenoides ornatus</i> Urban, 2005	0	0	0	2	7	3	0	4	1	0	17
<i>Anthrenoides paolae</i> Urban, 2005	0	2	7	3	1	0	6	1	7	0	27
<i>Anthrenoides petuniae</i> Urban, 2005	0	0	0	0	0	0	1	0	0	0	1
<i>Anthrenoides rodrigoii</i> Urban, 2005	0	0	0	0	0	1	0	0	0	2	3
<i>Anthrenoides</i> sp. 01	0	0	0	0	0	0	1	0	1	0	2
<i>Cephalurgus anomalus</i> Moure & Lucas de Oliveira, 1962	0	0	0	0	0	0	0	3	1	2	6
<i>Psaenythia annulata</i> Gerstaecker 1868	0	1	0	0	2	1	0	1	5	4	14
<i>Psaenythia bergii</i> Holmberg 1884	0	0	20	0	1	23	3	0	4	9	60
<i>Psaenythia capito</i> Gerstaecker, 1868	0	0	0	0	0	0	1	0	0	1	2
<i>Psaenythia collaris</i> Schrottky, 1906	0	2	5	1	1	3	2	2	5	0	21
<i>Psaenythia quadrifasciata</i> Friese, 1908	0	0	0	0	0	0	1	0	0	4	5
<i>Psaenythia</i> sp. 1	0	0	1	0	0	0	0	0	0	0	1
<i>Psaenythia</i> sp. 2	0	0	0	0	0	3	0	0	0	0	3
<i>Rhophitulus politus</i> (Schlindwein & Moure, 1998)	3	1	1	0	1	10	0	1	1	1	19
<i>Rhophitulus</i> sp. 1	0	2	0	6	4	0	0	0	0	0	12
<i>Rhophitulus</i> sp. 3	0	18	39	23	1	0	4	3	6	15	109
<i>Rhophitulus</i> sp. 4	0	0	0	1	0	0	0	1	4	4	10
<i>Rhophitulus</i> sp. 5	0	0	0	0	0	0	0	0	0	11	11
<i>Rhophitulus</i> sp. 6	0	0	0	0	1	1	0	0	0	0	2
<i>Rhophitulus</i> sp. 7	0	0	0	4	0	0	0	0	0	0	4
<i>Rhophitulus</i> sp. 8	0	0	0	1	0	1	0	19	0	1	22
<i>Rhophitulus</i> sp. 9	0	16	0	0	3	2	0	0	9	0	30
Apinae											
Bombini											
<i>Bombus</i> (<i>Fervidobombus</i>) <i>morio</i> (Swederus, 1787)	9	6	27	20	30	10	18	10	20	5	155
<i>Bombus</i> (<i>Fervidobombus</i>) <i>pauloensis</i> Friese, 1913	1	17	12	13	31	10	12	22	23	11	152
Euglossini											
<i>Eulaema</i> (<i>Apeulaema</i>) <i>nigrita</i> Lepeletier, 1841	0	0	0	0	0	0	0	1	0	0	1
Meliponini											
<i>Lestrimelitta sulina</i> Marchi & Melo, 2006	1	0	0	0	0	0	0	0	0	0	1
<i>Melipona</i> (<i>Eomelipona</i>) <i>bicolor</i> Lepeletier, 1836	0	0	0	1	0	0	1	0	0	0	2
<i>Melipona</i> (<i>Eomelipona</i>) <i>marginata</i> Lepeletier, 1836	0	32	1	5	0	0	4	4	0	1	47
<i>Melipona</i> (<i>Melipona</i>) <i>quadrifasciata</i> Lepeletier, 1836	0	5	7	25	5	5	5	1	16	0	69
<i>Melipona</i> (<i>Michmelia</i>) <i>scutellaris</i> Latreille, 1811	0	0	0	10	0	0	0	0	0	0	10
<i>Nannotrigona testaceicornis</i> (Lepeletier, 1836)	0	0	0	0	2	15	0	0	0	0	17
<i>Paratrigona subnuda</i> Moure, 1947	0	16	1	69	1	0	0	213	134	0	581
<i>Partamona helleri</i> (Friese, 1900)	22	21	0	2	0	1	0	5	9	0	60
<i>Plebeia droryana</i> (Friese, 1900)	20	21	0	0	8	0	0	2	4	0	55
<i>Plebeia emerina</i> (Friese, 1900)	42	5	0	28	30	23	61	1	0	39	229

<i>Plebeia remota</i> (Holmberg, 1903)	0	25	0	0	2	0	0	0	23	0	50
<i>Scaptotrigona bipunctata</i> (Lepeletier, 1836)	62	22	44	4	29	31	54	0	3	18	267
Bee species	1	2	3	4	5	6	7	8	9	10	Total
<i>Scaptotrigona postica</i> (Latreille, 1807)	0	0	0	0	0	2	0	0	0	0	2
<i>Schwarziana quadripunctata</i> (Lepeletier, 1836)	0	24	0	22	0	0	0	26	32	0	104
<i>Tetragonisca fiebrigi</i> (Schwarz, 1938)	11	52	59	51	21	43	11	64	28	4	344
<i>Trigona spinipes</i> (Fabricius, 1793)	31	77	103	45	43	25	81	69	55	44	573
Centridini											
<i>Centris</i> (<i>Centris</i>) <i>varia</i> (Erichson, 1849)	0	0	0	0	0	0	0	2	0	0	2
<i>Centris</i> (<i>Hemisiella</i>) <i>trigonoides</i> Lepeletier, 1841	1	0	2	4	5	0	0	3	0	0	15
<i>Centris</i> (<i>Hemisiella</i>) <i>tarsata</i> Smith, 1874	0	3	1	1	1	1	1	0	1	0	9
<i>Centris</i> (<i>Melacentris</i>) cfr. <i>lateritia</i> Friese, 1899	0	0	0	0	0	1	0	1	0	0	2
Emphorini											
<i>Melitoma segmentaria</i> (Fabricius, 1804)	0	3	7	8	0	4	2	6	10	0	40
<i>Ptilothrix relata</i> (Holmberg, 1903)	0	0	0	0	0	0	0	1	0	0	1
Eucerini											
<i>Gaesischia</i> (<i>Gaesischia</i>) <i>fulgurans</i> (Holmberg, 1903)	0	0	0	0	0	0	1	0	0	2	3
<i>Gaesischia</i> (<i>Gaesischia</i>) <i>undulata</i> Urban, 1989	0	0	0	0	0	0	2	0	0	0	2
<i>Gaesischia</i> (<i>Gaesischiopsis</i>) <i>aurea</i> Urban, 1968	0	0	0	0	0	0	0	0	0	1	1
<i>Gaesischia</i> (<i>Gaesischiopsis</i>) <i>flavoclypeata</i> Michener, LaBerge & Moure, 1955	0	0	0	0	0	0	0	0	0	2	2
<i>Melissodes</i> (<i>Ecplectica</i>) <i>nigroaenea</i> (Smith, 1854)	0	0	2	0	0	0	0	1	1	0	4
<i>Melissoptila aureocincta</i> Urban, 1968	0	0	9	15	4	0	6	0	2	2	38
<i>Melissoptila buzzii</i> Urban, 1998	0	2	4	10	0	2	6	0	1	3	28
<i>Melissoptila cnecomala</i> (Moure, 1944)	0	0	0	0	0	1	1	0	0	2	4
<i>Melissoptila fiebrigi</i> Brèthes, 1909	0	0	0	0	0	0	0	8	0	0	8
<i>Melissoptila larocai</i> Urban, 1998	0	0	0	0	0	0	2	0	4	3	9
<i>Melissoptila minarum</i> (Bertoni & Schrottkey, 1910)	0	0	0	0	0	0	3	0	0	1	4
<i>Melissoptila paraguayensis</i> (Brèthes, 1909)	0	0	0	0	10	0	0	0	7	1	18
<i>Melissoptila richardiae</i> Bertoni & Schrottkey, 1910	0	0	0	0	0	0	0	0	0	1	1
<i>Melissoptila setigera</i> Urban, 1998	0	1	20	12	0	9	2	0	1	8	53
<i>Melissoptila thoracica</i> (Smith, 1854)	0	2	2	0	15	15	4	1	0	15	54
<i>Melissoptila vulpecula</i> Bertoni & Schrottkey, 1910	0	0	0	0	0	0	0	0	0	8	8
<i>Peponapis fervens</i> (Smith, 1879)	0	0	0	0	1	0	0	0	0	0	1
<i>Thygater</i> (<i>Nectarodiaeta</i>) <i>paranaensis</i> Urban, 1967	0	2	0	0	0	0	1	0	0	0	3
<i>Thygater</i> (<i>Nectarodiaeta</i>) <i>sordidipenis</i> Moure, 1941	0	0	1	0	0	0	0	0	0	1	2
<i>Thygater</i> (<i>Thygater</i>) <i>analisis</i> (Lepeletier, 1841)	6	4	11	13	9	10	8	4	9	1	75
Exomalopsini											
<i>Exomalopsis</i> (<i>Diomalopsis</i>) <i>bicellularis</i> Michener & Moure, 1957	0	0	0	0	4	0	0	0	0	0	4
<i>Exomalopsis</i> (<i>Exomalopsis</i>) <i>analisis</i> Spinola 1853	1	0	3	5	14	10	5	1	1	0	40
<i>Exomalopsis</i> (<i>Exomalopsis</i>) <i>vernoniae</i> Schrottkey, 1909	3	0	0	0	1	12	0	2	4	0	22
<i>Exomalopsis</i> (<i>Exomalopsis</i>) sp. 01	9	2	1	4	3	14	1	3	7	0	44
<i>Exomalopsis</i> (<i>Exomalopsis</i>) sp. 02	6	0	1	1	2	7	0	2	1	0	20
<i>Exomalopsis</i> (<i>Phanomalopsis</i>) <i>aureosericea</i> Friese, 1899	0	3	0	5	0	5	0	2	3	0	18
<i>Exomalopsis</i> (<i>Phanomalopsis</i>) <i>perikalles</i> Silveira & Almeida, 2009	0	0	0	0	0	0	0	0	0	3	3
<i>Exomalopsis</i> (<i>Phanomalopsis</i>) <i>trifasciata</i> Brèthes, 1910	0	0	0	0	1	1	0	1	0	6	9
Nomadini											
<i>Brachynomada</i> sp. 01	0	1	0	0	0	0	0	0	0	1	2
<i>Brachynomada</i> sp. 03	0	0	1	0	0	0	0	0	0	0	1
<i>Leiopodus lacertinus</i> Smith, 1854	0	0	0	1	0	0	1	0	1	0	3
<i>Nomada</i> sp. 1	0	0	0	0	1	0	0	0	0	0	1
<i>Nomada</i> sp. 2	0	0	0	0	0	0	0	0	0	1	1
<i>Nomada</i> sp. 3	0	0	0	0	0	0	1	1	3	0	5
<i>Nomada</i> sp. 4	0	1	0	0	0	0	0	2	1	0	4
<i>Nomada</i> sp. 5	0	0	0	0	1	10	0	0	1	0	12

<i>Nomada</i> sp. 6	0	0	0	0	4	2	0	0	0	0	6
<i>Nomada</i> sp. 7	0	0	0	0	0	1	0	0	0	0	1
<i>Trophocleptria variolosa</i> Holmberg, 1886	0	0	0	0	0	0	2	0	0	0	2
Bee species	1	2	3	4	5	6	7	8	9	10	Total
<i>Trophocleptria</i> sp. 01	0	0	0	0	1	0	0	0	1	0	2
Osirini											
<i>Osiris</i> sp. 03	0	0	0	0	0	0	0	0	1	0	1
<i>Parepeolus minutus</i> Roig-Alsina, 1989	0	0	0	0	0	0	0	0	1	0	1
Tapinotaspidini											
<i>Arhysoceble picta</i> (Friese, 1899)	0	0	0	0	0	0	0	0	0	1	1
<i>Lanthanomelissa betinae</i> Urban, 1995	0	6	5	0	3	2	0	1	0	11	28
<i>Lanthanomelissa clementis</i> Urban, 1995	0	0	0	0	0	0	0	0	0	7	7
<i>Lophopedia nigrispinis</i> (Vachal, 1909)	0	1	0	0	1	0	0	1	1	0	4
<i>Lophopedia pygmaea</i> (Schrottky, 1902)	1	0	2	1	4	5	1	9	1	0	24
<i>Monoeca xanthopyga</i> Harter-Marques, Cunha & Moure, 2001	0	0	0	0	0	0	0	0	1	1	2
<i>Paratetrapedia fervida</i> (Smith, 1879)	0	0	0	0	0	0	1	0	2	4	7
<i>Trigonopedia</i> sp. 01	0	0	0	0	0	0	0	1	2	0	3
<i>Trigonopedia</i> sp. 03	0	0	0	0	0	0	0	3	2	0	5
Tetrapediini											
<i>Tetrapedia diversipes</i> Klug, 1810	0	2	2	3	1	3	4	1	10	0	26
Xylocopini											
<i>Ceratina</i> (<i>Ceratinula</i>) <i>melanochroa</i> (Moure, 1941)	0	1	0	6	10	9	1	7	0	6	40
<i>Ceratina</i> (<i>Ceratinula</i>) <i>oxalidis</i> Schrottky, 1907	2	3	0	4	7	1	0	4	1	2	24
<i>Ceratina</i> (<i>Ceratinula</i>) <i>sclerops</i> Schrottky, 1907	12	14	8	17	4	8	7	12	3	7	92
<i>Ceratina</i> (<i>Ceratinula</i>) sp. 01	0	0	2	0	0	2	0	0	0	0	4
<i>Ceratina</i> (<i>Ceratinula</i>) sp. 02	1	4	0	1	0	3	1	2	0	0	12
<i>Ceratina</i> (<i>Ceratinula</i>) sp. 03	0	0	0	0	0	0	0	0	0	1	1
<i>Ceratina</i> (<i>Ceratinula</i>) sp. 04	1	8	2	8	0	3	1	7	2	3	35
<i>Ceratina</i> (<i>Crewella</i>) <i>asuncionis</i> Strand, 1910	0	0	22	2	1	5	6	1	0	1	38
<i>Ceratina</i> (<i>Crewella</i>) sp. 03	8	2	15	32	23	36	18	6	7	11	158
<i>Ceratina</i> (<i>Neoclavicera</i>) <i>asunciana</i> Strand, 1910	0	0	1	0	0	0	2	0	0	2	5
<i>Ceratina</i> (<i>Neoclavicera</i>) <i>richardsoniae</i> Schrottky, 1909	19	2	1	3	8	14	20	2	2	29	100
<i>Ceratina</i> (<i>Neoclavicera</i>) <i>subcarinata</i> Roig-Alsina, 2013	0	0	0	0	0	0	7	0	1	7	15
<i>Ceratina</i> (<i>Neoclavicera</i>) <i>umbricosta</i> Roig-Alsina, 2013	0	0	0	0	3	0	1	0	0	1	5
<i>Ceratina</i> (<i>Rhysoceratina</i>) <i>canaliculata</i> Roig-Alsina, 2016	0	4	0	2	0	1	0	2	0	0	9
<i>Ceratina</i> (<i>Rhysoceratina</i>) <i>stilbonota</i> Moure, 1941	6	30	25	27	5	15	8	13	3	9	141
<i>Ceratina</i> (<i>Rhysoceratina</i>) <i>volitans</i> Schrottky, 1907	0	0	0	0	0	0	1	0	0	0	1
<i>Xylocopa</i> (<i>Neoxylocopa</i>) <i>augusti</i> Lepeletier, 1841	3	2	1	7	1	1	0	4	0	1	20
<i>Xylocopa</i> (<i>Neoxylocopa</i>) <i>frontalis</i> (Olivier, 1789)	1	1	2	2	0	0	1	1	4	2	14
<i>Xylocopa</i> (<i>Stenoxycopa</i>) <i>artifex</i> Smith, 1874	0	0	1	1	0	0	0	0	3	1	6
Colletinae											
Colletini											
<i>Colletes argentinus</i> Friese, 1908	0	0	0	4	3	0	3	2	3	0	15
<i>Colletes michenerianus</i> Moure, 1956	0	0	0	0	0	0	0	0	0	1	1
<i>Colletes rugicollis</i> Friese, 1900	0	0	0	0	3	2	1	0	0	0	6
Hylaeini											
<i>Hylaeus</i> (<i>Hylaeopsis</i>) <i>cecidonastes</i> Moure, 1972	0	0	0	0	0	0	0	1	0	0	1
<i>Hylaeus</i> sp. 01	0	1	2	1	3	2	3	4	4	0	20
<i>Hylaeus</i> sp. 02	7	3	3	5	6	1	3	0	0	6	34
<i>Hylaeus</i> sp. 03	2	1	0	0	1	2	0	0	2	0	8
<i>Hylaeus</i> sp. 04	0	0	0	0	1	2	0	0	0	0	3
<i>Hylaeus</i> sp. 05	18	1	3	1	1	2	3	5	8	0	42
<i>Hylaeus</i> sp. 06	1	1	2	5	6	0	0	5	7	1	28
<i>Hylaeus</i> sp. 07	0	0	0	0	0	0	0	1	1	0	2
<i>Hylaeus</i> sp. 08	0	2	0	7	1	1	2	6	5	1	25

<i>Hylaeus</i> sp. 09	0	0	0	0	2	0	0	0	0	0	2
Neopasephacini											
<i>Belopria zonata</i> Moure, 1956	0	0	0	0	0	0	1	0	0	0	1
<i>Belopria</i> sp. 02	0	0	0	0	0	0	0	0	1	0	1
Bee species	1	2	3	4	5	6	7	8	9	10	Total
<i>Bicolletes tauraphilus</i> Laroca & Almeida, 2005	0	0	0	0	2	1	0	2	0	0	5
<i>Ctenosibyne cingulata</i> (Moure, 1956)	0	1	0	3	0	0	0	0	0	0	4
<i>Eulonchopria psanythioides</i> Brêthes, 1909	0	0	1	0	0	0	0	0	0	0	1
<i>Tetraglossula anthracina</i> (Michener, 1989)	0	5	5	10	0	10	10	6	7	24	77
Xeromelissini											
<i>Chilicola (Oediscelisca)</i> sp. 01	0	0	1	1	0	0	0	0	0	6	8
<i>Chilicola (Oediscelisca)</i> sp. 02	0	2	0	0	1	0	0	1	0	1	5
<i>Chilicola (Prosopoides)</i> cfr. <i>prosopoides</i> (Ducke, 1907)	0	0	1	0	2	2	0	12	3	1	21
Halictinae											
Augochlorini											
<i>Augochlora (Augochlora) caerulior</i> Cockerell, 1900	0	3	1	4	9	8	1	0	2	1	29
<i>Augochlora (Augochlora) cydippe</i> (Schrottky, 1910)	0	3	0	3	7	10	4	3	3	5	38
<i>Augochlora (Augochlora) daphnis</i> Smith, 1853	110	59	56	59	48	50	51	44	17	53	547
<i>Augochlora (Augochlora) nausicaa</i> (Schrottky, 1909)	16	1	4	5	1	10	0	2	2	0	41
<i>Augochlora (Augochlora) perimelas</i> Cockerell, 1900	0	0	0	0	0	0	3	1	1	0	5
<i>Augochlora (Augochlora) phoemonoe</i> (Schrottky, 1909)	10	14	6	2	12	15	9	2	2	2	74
<i>Augochlora (Augochlora) seitzii</i> Cockerell, 1929	13	0	1	0	2	0	1	0	0	0	17
<i>Augochlora (Augochlora) esox</i> (Vachal, 1911)	1	1	0	0	0	0	0	0	0	0	2
<i>Augochlora (Augochlora) genalis</i> Lepeco & Gonçalves, 2020	0	0	0	0	0	0	2	0	0	0	2
<i>Augochlora (Oxystoglossella) iphigenia</i> Holmberg, 1886	0	45	54	6	31	9	29	37	55	10	276
<i>Augochlora (Oxystoglossella) morrae</i> (Strand, 1910)	0	1	1	6	4	6	5	4	45	1	73
<i>Augochlorella ephyra</i> (Schrottky, 1910)	54	7	0	8	3	2	0	3	0	1	78
<i>Augochlorella iopoecila</i> Moure, 1950	0	1	1	0	0	8	0	0	1	4	15
<i>Augochlorella urania</i> (Smith, 1853)	25	4	5	4	2	0	6	1	1	1	49
<i>Augochlorella acarinata</i> Coelho, 2004	0	0	0	0	0	3	0	0	0	0	3
<i>Augochloropsis (Augochloropsis) brachycephala</i> Moure, 1943	0	19	9	9	36	47	7	4	37	1	169
<i>Augochloropsis (Augochloropsis)</i> sp. 11	0	0	0	1	0	0	0	0	0	0	1
<i>Augochloropsis (Glyptobasia) chloera</i> (Moure, 1940)	0	0	0	0	1	0	0	0	1	0	2
<i>Augochloropsis (Paraugochloropsis) cleopatra</i> (Schrottky, 1902)	0	0	17	2	0	0	0	0	0	2	21
<i>Augochloropsis (Paraugochloropsis) cupreola</i> (Cockerell, 1900)	7	0	3	4	9	1	8	5	1	4	42
<i>Augochloropsis (Paraugochloropsis) iris</i> (Schrottky, 1902)	0	0	0	0	0	0	0	0	0	2	2
<i>Augochloropsis (Paraugochloropsis) multiplex</i> (Vachal, 1903)	0	0	14	5	1	1	10	0	1	11	43
<i>Augochloropsis (Paraugochloropsis) sparsilis</i> (Vachal, 1903)	0	0	37	6	17	6	19	0	0	17	102
<i>Augochloropsis (Paraugochloropsis)</i> sp. 02	0	4	2	1	2	4	1	1	0	1	16
<i>Augochloropsis (Paraugochloropsis)</i> sp. 03	0	3	1	3	0	2	4	4	9	1	27
<i>Augochloropsis (Paraugochloropsis)</i> sp. 05	1	5	2	4	7	1	3	6	0	9	38
<i>Augochloropsis (Paraugochloropsis)</i> sp. 09	0	15	10	3	0	2	1	2	17	1	51
<i>Augochloropsis (Paraugochloropsis)</i> sp. 10	0	0	1	0	0	0	0	0	1	0	2
<i>Augochloropsis (Paraugochloropsis)</i> sp. 11	0	0	0	1	0	0	0	0	0	0	1
<i>Augochloropsis (Paraugochloropsis)</i> sp. 12	0	0	0	0	0	0	0	0	0	1	1
<i>Augochloropsis (Paraugochloropsis)</i> sp. 13	0	1	0	0	0	0	0	0	0	1	2
<i>Augochloropsis (Paraugochloropsis)</i> sp. 14	0	0	0	0	0	0	0	0	0	1	1
<i>Augochloropsis (Paraugochloropsis)</i> sp. 15	0	0	0	0	1	1	0	0	0	0	2
<i>Ceratalictus psoraspis</i> (Vachal, 1911)	0	0	1	0	0	0	1	0	1	5	8
<i>Corynurella cognata</i> Gonçalves, 2010	0	0	0	0	0	0	0	0	0	1	1
<i>Halictillus loureiroi</i> (Moure, 1941)	0	2	0	0	0	3	0	11	0	0	16
<i>Megommation insigne</i> (Smith, 1853)	0	0	0	0	1	0	1	0	0	0	2
<i>Neocorynura atromarginata</i> (Cockerell, 1901)	0	0	0	3	0	0	0	0	0	3	6
<i>Neocorynura codion</i> (Vachal, 1904)	60	14	24	19	25	80	39	7	21	10	299
<i>Neocorynura dilutipes</i> (Vachal, 1904)	7	12	7	13	13	1	4	1	2	13	73

<i>Dialictus</i> sp. 19	0	0	0	0	0	0	1	0	0	0	1
Sphecodini											
<i>Austrosphcodes brasiliensis</i> Schrottky, 1910	0	0	0	0	0	0	0	0	1	0	1
<i>Austrosphcodes</i> sp. 01	0	0	0	0	4	1	0	0	0	0	5
<i>Austrosphcodes</i> sp. 03	0	0	0	1	0	0	0	0	0	0	1
<i>Melissocleptis capriciosa</i> (Schrottky, 1906)	0	1	1	0	0	0	0	0	0	0	2
Bee species	1	2	3	4	5	6	7	8	9	10	Total
<i>Microsphcodes russeiclypeatus</i> (Sakagami & Moure, 1962)	0	0	0	0	0	0	1	1	0	0	2
<i>Nesosphcodes depressus</i> Gonçalves, 2021	0	0	1	0	0	0	0	0	0	0	1
Megachilinae											
Anthidiini											
<i>Anthidium manicatum</i> (Linnaeus, 1758)	0	0	2	0	0	1	0	1	0	0	4
<i>Anthodioctes claudii</i> Urban, 1999	2	0	0	0	1	0	0	0	0	0	3
<i>Anthodioctes megachiloides</i> Holmberg, 1903	2	0	0	0	0	0	0	0	0	0	2
<i>Ctenanthidium gracile</i> Urban, 1993	0	0	0	0	0	0	1	0	0	0	1
<i>Dicranthidium</i> aff. <i>soniae</i> Urban, 1993	0	0	0	0	0	0	0	3	0	0	3
<i>Dicranthidium</i> sp. 01	0	0	0	0	0	0	0	1	0	0	1
<i>Epanthidium autumnale</i> (Schrottky, 1909)	0	1	0	0	0	0	0	0	2	0	3
<i>Epanthidium nectarinioides</i> (Schrottky, 1902)	0	0	0	0	3	1	1	0	0	0	5
<i>Hypanthidium divaricatum</i> (Smith, 1854)	0	3	0	0	0	6	0	2	2	0	13
<i>Moureanthidium paranaense</i> Urban, 1995	0	1	1	2	1	4	0	0	0	0	9
<i>Moureanthidium subarenarium</i> (Schwarz, 1933)	0	1	1	0	0	0	0	0	0	0	2
Megachilini											0
<i>Coelioxys (Acrocoelioxys) tolteca</i> Cresson, 1878	1	2	2	4	11	9	1	2	3	4	39
<i>Coelioxys (Cyrtocoelioxys)</i> sp. 01	0	0	2	1	3	2	0	2	0	2	12
<i>Coelioxys (Cyrtocoelioxys)</i> sp. 02	0	0	1	0	0	0	0	0	0	0	1
<i>Coelioxys (Cyrtocoelioxys)</i> sp. 03	0	0	0	0	0	0	0	1	0	0	1
<i>Coelioxys (Glyptocoelioxys) cerasiopleura</i> Holmberg, 1903	0	0	2	0	0	0	0	0	0	2	4
<i>Coelioxys (Glyptocoelioxys) chacoensis</i> Holmberg, 1903	0	0	0	0	0	0	0	0	0	1	1
<i>Coelioxys (Glyptocoelioxys)</i> sp. 01	0	0	0	0	0	0	0	0	1	0	1
<i>Coelioxys (Neocoelioxys) praetextata</i> Haliday, 1836	0	0	0	0	0	0	0	0	0	1	1
<i>Coelioxys (Neocoelioxys)</i> sp. 01	0	1	0	1	0	0	0	0	0	0	2
<i>Megachile (Acentron) lentifera</i> Vachal, 1909	0	0	0	0	0	0	3	0	0	2	5
<i>Megachile (Acentron)</i> sp. 01	0	0	0	0	0	0	0	0	0	3	3
<i>Megachile (Austromegachile) facialis</i> Vachal, 1909	0	0	0	0	2	0	0	0	0	0	2
<i>Megachile (Austromegachile) susurrans</i> Haliday, 1836	8	1	4	1	10	9	7	5	1	3	49
<i>Megachile (Austromegachile) trigonasps</i> Schrottky, 1913	0	1	0	0	1	0	0	1	0	0	3
<i>Megachile (Austromegachile)</i> sp. 01	1	0	0	0	1	0	0	0	0	0	2
<i>Megachile (Chrysosarus)</i> cfr. <i>tuberculifera</i> Schrottky, 1913	0	1	0	1	1	2	1	0	2	2	10
<i>Megachile (Chrysosarus)</i> sp. 01	0	0	3	2	0	1	2	1	0	2	11
<i>Megachile (Leptorachina)</i> sp. 01	0	0	1	0	0	0	0	0	0	0	1
<i>Megachile (Leptorachis) aetheria</i> Mitchell, 1930	0	0	4	4	1	2	1	0	1	0	13
<i>Megachile (Leptorachis) paulistana</i> Schrottky, 1902	0	0	0	0	1	1	1	0	3	0	6
<i>Megachile (Leptorachis)</i> sp. 01	0	0	0	0	0	0	0	1	1	0	2
<i>Megachile (Melanosarus)</i> aff. <i>brasiliensis</i> Dalla Torre, 1896	0	1	0	5	0	0	0	0	0	0	6
<i>Megachile (Moureapis) apicipennis</i> Schrottky, 1902	9	10	8	20	38	26	22	11	9	10	163
<i>Megachile (Moureapis) maculata</i> Smith, 1853	0	4	1	0	0	3	4	0	0	0	12
<i>Megachile (Moureapis) nigropilosa</i> Schrottky, 1902	0	0	3	0	0	1	0	1	0	3	8
<i>Megachile (Pseudocentron) curvipes</i> Smith, 1853	0	0	7	0	1	7	5	0	1	0	21
<i>Megachile (Tylomegachile) orba</i> Schrottky, 1913	0	0	0	0	0	0	0	0	0	2	2
<i>Megachile</i> sp. 01	0	0	1	0	0	1	0	0	0	0	2
<i>Megachile</i> sp. 02	0	3	0	0	0	2	0	0	1	1	7

Online Resource 5. Bee species and their functional traits in the city of Curitiba, Southern Brazil. Behavior: (SO) solitary, (P_E) primitively eusocial, (A_E) highly eusocial, (CL) cleptoparasitic. Nidification: (G) ground, (A) above ground. Trophic specialization: (S) specialist, (GE) generalist. (NA) not specified.

Bee species	Behavior	Nesting site	Feeding habit
Andreninae			
Calliopsini			
<i>Acamptopoeum prini</i> (Holmberg, 1884)	SO	G	GE
<i>Callonychium</i> (<i>Callonychium</i>) <i>petuniae</i> Cure & Wittmann, 1990	SO	G	S
Protandrenini			
<i>Anthrenoides admirabilis</i> Urban, 2005	SO	G	S
<i>Anthrenoides alvarengai</i> Urban, 2007	SO	G	S
<i>Anthrenoides antonii</i> Urban, 2005	SO	G	S
<i>Anthrenoides araucariae</i> Urban, 2005	SO	G	S
<i>Anthrenoides corrugatus</i> Urban, 2005	SO	G	S
<i>Anthrenoides larocai</i> Urban, 2005	SO	G	S
<i>Anthrenoides meridionalis</i> (Schrottky, 1906)	SO	G	S
<i>Anthrenoides ornatus</i> Urban, 2005	SO	G	S
<i>Anthrenoides paolae</i> Urban, 2005	SO	G	S
<i>Anthrenoides petuniae</i> Urban, 2005	SO	G	S
<i>Anthrenoides rodrigo</i> Urban, 2005	SO	G	S
<i>Anthrenoides</i> sp. 01	SO	G	S
<i>Cephalurgus anomalus</i> Moure & Lucas de Oliveira, 1962	SO	G	S
<i>Psaenythia annulata</i> Gerstaecker 1868	SO	G	GE
<i>Psaenythia bergii</i> Holmberg 1884	SO	G	GE
<i>Psaenythia capito</i> Gerstaecker, 1868	SO	G	GE
<i>Psaenythia collaris</i> Schrottky, 1906	SO	G	GE
<i>Psaenythia quadrifasciata</i> Friese, 1908	SO	G	GE
<i>Psaenythia</i> sp. 1	SO	G	GE
<i>Psaenythia</i> sp. 2	SO	G	GE
<i>Rhophitulus politus</i> (Schlindwein & Moure, 1998)	SO	G	S
<i>Rhophitulus</i> sp. 1	SO	G	S
<i>Rhophitulus</i> sp. 3	SO	G	S
<i>Rhophitulus</i> sp. 4	SO	G	S
<i>Rhophitulus</i> sp. 5	SO	G	S
<i>Rhophitulus</i> sp. 6	SO	G	S
<i>Rhophitulus</i> sp. 7	SO	G	S
<i>Rhophitulus</i> sp. 8	SO	G	S
<i>Rhophitulus</i> sp. 9	SO	G	S
Apinae			
Bombini			
<i>Bombus</i> (<i>Fervidobombus</i>) <i>morio</i> (Swederus, 1787)	P_E	A	GE
<i>Bombus</i> (<i>Fervidobombus</i>) <i>pauloensis</i> Friese, 1913	P_E	A	GE
Euglossini			
<i>Eulaema</i> (<i>Apeulaema</i>) <i>nigrita</i> Lepeletier, 1841	SO	G	GE
Meliponini			
<i>Lestrimelitta sulina</i> Marchi & Melo, 2006	A_E	A	GE
<i>Melipona</i> (<i>Eomelipona</i>) <i>bicolor</i> Lepeletier, 1836	A_E	A	GE
<i>Melipona</i> (<i>Eomelipona</i>) <i>marginata</i> Lepeletier, 1836	A_E	A	GE
<i>Melipona</i> (<i>Melipona</i>) <i>quadrifasciata</i> Lepeletier, 1836	A_E	A	GE

Bee species	Behavior	Nesting site	Feeding habit
<i>Melipona (Michmelia) scutellaris</i> Latreille, 1811	A_E	A	GE
<i>Nannotrigona testaceicornis</i> (Lepeletier, 1836)	A_E	A	GE
<i>Paratrigona subnuda</i> Moure, 1947	A_E	G	GE
<i>Partamona helleri</i> (Fries, 1900)	A_E	A	GE
<i>Plebeia droryana</i> (Fries, 1900)	A_E	A	GE
<i>Plebeia emerina</i> (Fries, 1900)	A_E	A	GE
<i>Plebeia remota</i> (Holmberg, 1903)	A_E	A	GE
<i>Scaptotrigona bipunctata</i> (Lepeletier, 1836)	A_E	A	GE
<i>Scaptotrigona postica</i> (Latreille, 1807)	A_E	A	GE
<i>Schwarziana quadripunctata</i> (Lepeletier, 1836)	A_E	G	GE
<i>Tetragonisca fiebrigi</i> (Schwarz, 1938)	A_E	A	GE
<i>Trigona spinipes</i> (Fabricius, 1793)	A_E	A	GE
Centridini			
<i>Centris (Centris) varia</i> (Erichson, 1849)	SO	G	S
<i>Centris (Hemisiella) trigonoides</i> Lepeletier, 1841	SO	A	S
<i>Centris (Hemisiella) tarsata</i> Smith, 1874	SO	A	S
<i>Centris (Melacentris) cfr. lateritia</i> Fries, 1899	SO	G	S
Emphorini			
<i>Melitoma segmentaria</i> (Fabricius, 1804)	SO	G	S
<i>Ptilothrix relata</i> (Holmberg, 1903)	SO	G	S
Eucerini			
<i>Gaesischia (Gaesischia) fulgurans</i> (Holmberg, 1903)	SO	G	S
<i>Gaesischia (Gaesischia) undulata</i> Urban, 1989	SO	G	S
<i>Gaesischia (Gaesischiopsis) aurea</i> Urban, 1968	SO	G	S
<i>Gaesischia (Gaesischiopsis) flavoclypeata</i> Michener, LaBerge & Moure, 1955	SO	G	S
<i>Melissodes (Ecplectica) nigroaenea</i> (Smith, 1854)	SO	G	GE
<i>Melissoptila aureocincta</i> Urban, 1968	SO	G	S
<i>Melissoptila buzzii</i> Urban, 1998	SO	G	S
<i>Melissoptila cnecomala</i> (Moure, 1944)	SO	G	S
<i>Melissoptila fiebrigi</i> Brèthes, 1909	SO	G	S
<i>Melissoptila larocai</i> Urban, 1998	SO	G	S
<i>Melissoptila minarum</i> (Bertoni & Schrottky, 1910)	SO	G	S
<i>Melissoptila paraguayensis</i> (Brèthes, 1909)	SO	G	S
<i>Melissoptila richardiae</i> Bertoni & Schrottky, 1910	SO	G	S
<i>Melissoptila setigera</i> Urban, 1998	SO	G	S
<i>Melissoptila thoracica</i> (Smith, 1854)	SO	G	S
<i>Melissoptila vulpecula</i> Bertoni & Schrottky, 1910	SO	G	S
<i>Peponapis fervens</i> (Smith, 1879)	SO	G	S
<i>Thygater (Nectarodiaeta) paranaensis</i> Urban, 1967	SO	G	GE
<i>Thygater (Nectarodiaeta) sordidipenis</i> Moure, 1941	SO	G	GE
<i>Thygater (Thygater) analis</i> (Lepeletier, 1841)	SO	G	GE
Exomalopsini			
<i>Exomalopsis (Diomalopsis) bicellularis</i> Michener & Moure, 1957	SO	G	GE
<i>Exomalopsis (Exomalopsis) analis</i> Spinola 1853	SO	G	GE
<i>Exomalopsis (Exomalopsis) vernoniae</i> Schrottky, 1909	SO	G	GE
<i>Exomalopsis (Exomalopsis) sp. 01</i>	SO	G	GE
<i>Exomalopsis (Exomalopsis) sp. 02</i>	SO	G	GE
<i>Exomalopsis (Phanomalopsis) aureosericea</i> Fries, 1899	SO	G	GE
<i>Exomalopsis (Phanomalopsis) perikalles</i> Silveira & Almeida, 2009	SO	G	GE

Bee species	Behavior	Nesting site	Feeding habit
<i>Exomalopsis (Phanomalopsis) trifasciata</i> Brèthes, 1910	SO	G	GE
Nomadini			
<i>Brachynomada</i> sp. 01	CL	NA	NA
<i>Brachynomada</i> sp. 03	CL	NA	NA
<i>Leiopodus lacertinus</i> Smith, 1854	CL	NA	NA
<i>Nomada</i> sp. 1	CL	NA	NA
<i>Nomada</i> sp. 2	CL	NA	NA
<i>Nomada</i> sp. 3	CL	NA	NA
<i>Nomada</i> sp. 4	CL	NA	NA
<i>Nomada</i> sp. 5	CL	NA	NA
<i>Nomada</i> sp. 6	CL	NA	NA
<i>Nomada</i> sp. 7	CL	NA	NA
<i>Pseudepeolus angustatus</i> (Moure, 1954)	CL	NA	NA
<i>Trophocleptria variolosa</i> Holmberg, 1886	CL	NA	NA
<i>Trophocleptria</i> sp. 01	CL	NA	NA
Osirini			
<i>Osiris</i> sp. 03	CL	NA	NA
<i>Parepeolus minutus</i> Roig-Alsina, 1989	CL	NA	NA
Tapinotaspidini			
<i>Arhysoceble picta</i> (Friese, 1899)	SO	G	GE
<i>Lanthanomalissa betinae</i> Urban, 1995	SO	G	S
<i>Lanthanomalissa clementis</i> Urban, 1995	SO	G	S
<i>Lophopedia nigrispinis</i> (Vachal, 1909)	SO	A	S
<i>Lophopedia pygmaea</i> (Schrottky, 1902)	SO	A	S
<i>Monoeca xanthopyga</i> Harter-Marques, Cunha & Moure, 2001	SO	G	S
<i>Paratetrapedia fervida</i> (Smith, 1879)	SO	A	S
<i>Trigonopedia</i> sp. 01	SO	G	S
<i>Trigonopedia</i> sp. 03	SO	G	S
Tetrapediini			
<i>Tetrapedia diversipes</i> Klug, 1810	SO	A	GE
Xylocopini			
<i>Ceratina (Ceratinula) melanochoa</i> (Moure, 1941)	SO	A	GE
<i>Ceratina (Ceratinula) oxalidis</i> Schrottky, 1907	SO	A	GE
<i>Ceratina (Ceratinula) sclerops</i> Schrottky, 1907	SO	A	GE
<i>Ceratina (Ceratinula)</i> sp. 01	SO	A	GE
<i>Ceratina (Ceratinula)</i> sp. 02	SO	A	GE
<i>Ceratina (Ceratinula)</i> sp. 03	SO	A	GE
<i>Ceratina (Ceratinula)</i> sp. 04	SO	A	GE
<i>Ceratina (Crewella) asuncionis</i> Strand, 1910	SO	A	GE
<i>Ceratina (Crewella)</i> sp. 03	SO	A	GE
<i>Ceratina (Neoclavicera) asunciana</i> Strand, 1910	SO	A	GE
<i>Ceratina (Neoclavicera) richardsoniae</i> Schrottky, 1909	SO	A	GE
<i>Ceratina (Neoclavicera) subcarinata</i> Roig-Alsina, 2013	SO	A	GE
<i>Ceratina (Neoclavicera) umbricosta</i> Roig-Alsina, 2013	SO	A	GE
<i>Ceratina (Rhysoceratina) canaliculata</i> Roig-Alsina, 2016	SO	A	GE
<i>Ceratina (Rhysoceratina) stilbonota</i> Moure, 1941	SO	A	GE
<i>Ceratina (Rhysoceratina) volitans</i> Schrottky, 1907	SO	A	GE
<i>Xylocopa (Neoxylocopa) augusti</i> Lepeletier, 1841	SO	A	GE
<i>Xylocopa (Neoxylocopa) frontalis</i> (Olivier, 1789)	SO	A	GE
<i>Xylocopa (Stenoxycopa) artifex</i> Smith, 1874	SO	A	GE

Bee species	Behavior	Nesting site	Feeding habit
Colletinae			
Colletini			
<i>Colletes argentinus</i> Friese, 1908	SO	G	GE
<i>Colletes michenerianus</i> Moure, 1956	SO	G	GE
<i>Colletes rugicollis</i> Friese, 1900	SO	G	GE
Hylaeini			
<i>Hylaeus (Hylaeopsis) cecidonastes</i> Moure, 1972	SO	A	GE
<i>Hylaeus</i> sp. 01	SO	A	GE
<i>Hylaeus</i> sp. 02	SO	A	GE
<i>Hylaeus</i> sp. 03	SO	A	GE
<i>Hylaeus</i> sp. 04	SO	A	GE
<i>Hylaeus</i> sp. 05	SO	A	GE
<i>Hylaeus</i> sp. 06	SO	A	GE
<i>Hylaeus</i> sp. 07	SO	A	GE
<i>Hylaeus</i> sp. 08	SO	A	GE
<i>Hylaeus</i> sp. 09	SO	A	GE
Neopasephaeini			
<i>Belopria zonata</i> Moure, 1956	SO	G	S
<i>Belopria</i> sp. 02	SO	G	S
<i>Bicolletes tauraphilus</i> Laroca & Almeida, 2005	SO	G	S
<i>Ctenosibyne cingulata</i> (Moure, 1956)	SO	G	S
<i>Eulonchopria psauenythioides</i> Brèthes, 1909	SO	G	S
<i>Tetraglossula anthracina</i> (Michener, 1989)	SO	G	S
Xeromelissini			
<i>Chilicola (Oediscelisca)</i> sp. 01	SO	A	GE
<i>Chilicola (Oediscelisca)</i> sp. 02	SO	A	GE
<i>Chilicola (Prosopoides)</i> cfr. <i>prosopoides</i> (Ducke, 1907)	SO	A	GE
Halictinae			
Augochlorini			
<i>Augochlora (Augochlora) caerulior</i> Cockerell, 1900	P_E	A	GE
<i>Augochlora (Augochlora) cydippe</i> (Schrottky, 1910)	P_E	A	GE
<i>Augochlora (Augochlora) daphnis</i> Smith, 1853	P_E	A	GE
<i>Augochlora (Augochlora) nausicaa</i> (Schrottky, 1909)	P_E	A	GE
<i>Augochlora (Augochlora) perimelas</i> Cockerell, 1900	P_E	A	GE
<i>Augochlora (Augochlora) phoemonoe</i> (Schrottky, 1909)	P_E	A	GE
<i>Augochlora (Augochlora) seitzii</i> Cockerell, 1929	P_E	A	GE
<i>Augochlora (Augochlora) esox</i> (Vachal, 1911)	P_E	A	GE
<i>Augochlora (Augochlora) genalis</i> Lepeco & Gonçalves, 2020	P_E	A	GE
<i>Augochlora (Oxystoglossella) iphigenia</i> Holmberg, 1886	P_E	G	GE
<i>Augochlora (Oxystoglossella) morrae</i> (Strand, 1910)	P_E	G	GE
<i>Augochlorella ephyra</i> (Schrottky, 1910)	P_E	G	GE
<i>Augochlorella iopoecila</i> Moure, 1950	P_E	G	GE
<i>Augochlorella urania</i> (Smith, 1853)	P_E	G	GE
<i>Augochlorella acarinata</i> Coelho, 2004	P_E	G	GE
<i>Augochloropsis (Augochloropsis) brachycephala</i> Moure, 1943	SO	G	GE
<i>Augochloropsis (Augochloropsis)</i> sp. 11	SO	G	GE
<i>Augochloropsis (Glyptobasia) chloera</i> (Moure, 1940)	SO	G	GE
<i>Augochloropsis (Paraugochloropsis) cleopatra</i> (Schrottky, 1902)	SO	G	GE
<i>Augochloropsis (Paraugochloropsis) cupreola</i> (Cockerell, 1900)	SO	G	GE
<i>Augochloropsis (Paraugochloropsis) iris</i> (Schrottky, 1902)	SO	G	GE

Bee species	Behavior	Nesting site	Feeding habit
<i>Augochloropsis (Paraugochloropsis) multiplex</i> (Vachal, 1903)	SO	G	GE
<i>Augochloropsis (Paraugochloropsis) sparsilis</i> (Vachal, 1903)	SO	G	GE
<i>Augochloropsis (Paraugochloropsis)</i> sp. 02	SO	G	GE
<i>Augochloropsis (Paraugochloropsis)</i> sp. 03	SO	G	GE
<i>Augochloropsis (Paraugochloropsis)</i> sp. 05	SO	G	GE
<i>Augochloropsis (Paraugochloropsis)</i> sp. 09	SO	G	GE
<i>Augochloropsis (Paraugochloropsis)</i> sp. 10	SO	G	GE
<i>Augochloropsis (Paraugochloropsis)</i> sp. 11	SO	G	GE
<i>Augochloropsis (Paraugochloropsis)</i> sp. 12	SO	G	GE
<i>Augochloropsis (Paraugochloropsis)</i> sp. 13	SO	G	GE
<i>Augochloropsis (Paraugochloropsis)</i> sp. 14	SO	G	GE
<i>Augochloropsis (Paraugochloropsis)</i> sp. 15	SO	G	GE
<i>Ceratalictus psoraspis</i> (Vachal, 1911)	SO	G	GE
<i>Corynurella cognata</i> Gonçalves, 2010	SO	G	GE
<i>Halictillus loureiroi</i> (Moure, 1941)	SO	G	GE
<i>Megommation insigne</i> (Smith, 1853)	SO	G	GE
<i>Neocorynura atromarginata</i> (Cockerell, 1901)	SO	G	GE
<i>Neocorynura codion</i> (Vachal, 1904)	SO	G	GE
<i>Neocorynura dilutipes</i> (Vachal, 1904)	SO	G	GE
<i>Neocorynura hebe</i> Gonçalves, 2019	SO	G	GE
<i>Neocorynura iopodion</i> (Vachal, 1904)	SO	G	GE
<i>Neocorynura laevistriata</i> Gonçalves, 2019	SO	G	GE
<i>Neocorynura roxane</i> (Schrottky, 1909)	SO	G	GE
<i>Paroxystoglossa andromache</i> (Schrottky, 1909)	SO	G	GE
<i>Paroxystoglossa brachycera</i> Moure, 1960	SO	G	GE
<i>Paroxystoglossa jocasta</i> (Schrottky, 1910)	SO	G	GE
<i>Paroxystoglossa spiloptera</i> Moure, 1960	SO	G	GE
<i>Pseudaugochlora graminea</i> (Fabricius, 1804)	SO	G	GE
<i>Pseudaugochlora indistincta</i> Almeida, 2008	SO	G	GE
<i>Rhectomia pumilla</i> Moure, 1947	SO	G	GE
<i>Rhinocorynura vernoniae</i> (Schrottky, 1914)	P_E	G	GE
<i>Temnosoma</i> sp. 01	CL	NA	NA
<i>Temnosoma</i> sp. 02	CL	NA	NA
<i>Temnosoma</i> sp. 03	CL	NA	NA
<i>Thectochlora alaris</i> (Vachal, 1904)	SO	G	GE
Caenohalictini			
<i>Agapostemon (Notagapostemon) semimelleus</i> Cockerell, 1900	SO	G	GE
<i>Caenohalictus curticeps</i> (Vachal, 1903)	SO	G	GE
<i>Caenohalictus incertus</i> (Schrottky, 1902)	SO	G	GE
<i>Caenohalictus</i> cfr. <i>palumbes</i> (Vachal, 1903)	SO	G	GE
<i>Caenohalictus palumbes</i> (Vachal, 1903)	SO	G	GE
<i>Caenohalictus tessellatus</i> (Moure, 1940)	SO	G	GE
<i>Caenohalictus</i> sp. 01	SO	G	GE
<i>Habralictus canaliculatus</i> Moure, 1941	SO	G	GE
<i>Habralictus macrospilophorus</i> Moure, 1941	SO	G	GE
<i>Oragapostemon divaricatus</i> (Vachal, 1904)	SO	G	GE
<i>Pseudagapostemon (Neagapostemon) cyanomelas</i> Cure, 1989	SO	G	GE
<i>Pseudagapostemon (Pseudagapostemon) cyaneus</i> Moure & Sakagami, 1984	SO	G	GE

Bee species	Behavior	Nesting site	Feeding habit
<i>Pseudagapostemon (Pseudagapostemon) pruinosus</i> Moure & Sakagami, 1984	SO	G	GE
Halictini			
<i>Dialictus nanus</i> (Smith, 1879)	P_E	G	GE
<i>Dialictus opacus</i> (Moure, 1940)	P_E	G	GE
<i>Dialictus phaedrus</i> (Schrottky, 1910)	P_E	G	GE
<i>Dialictus rostratus</i> (Moure, 1947)	P_E	G	GE
<i>Dialictus</i> sp. 01	P_E	G	GE
<i>Dialictus</i> sp. 02	P_E	G	GE
<i>Dialictus</i> sp. 03	P_E	G	GE
<i>Dialictus</i> sp. 04	P_E	G	GE
<i>Dialictus</i> sp. 05	P_E	G	GE
<i>Dialictus</i> sp. 06	P_E	G	GE
<i>Dialictus</i> sp. 07	P_E	G	GE
<i>Dialictus</i> sp. 08	P_E	G	GE
<i>Dialictus</i> sp. 09	P_E	G	GE
<i>Dialictus</i> sp. 11	P_E	G	GE
<i>Dialictus</i> sp. 12	P_E	G	GE
<i>Dialictus</i> sp. 13	P_E	G	GE
<i>Dialictus</i> sp. 14	P_E	G	GE
<i>Dialictus</i> sp. 15	P_E	G	GE
<i>Dialictus</i> sp. 16	P_E	G	GE
<i>Dialictus</i> sp. 17	P_E	G	GE
<i>Dialictus</i> sp. 18	P_E	G	GE
<i>Dialictus</i> sp. 19	P_E	G	GE
Sphecodini			
<i>Austrosphcodes brasiliensis</i> Schrottky, 1910	CL	NA	NA
<i>Austrosphcodes</i> sp. 01	CL	NA	NA
<i>Austrosphcodes</i> sp. 03	CL	NA	NA
<i>Melissocleptis capriciosa</i> (Schrottky, 1906)	CL	NA	NA
<i>Microsphcodes russeiclypeatus</i> (Sakagami & Moure, 1962)	CL	NA	NA
<i>Nesosphcodes depressus</i> Gonçalves, 2021	CL	NA	NA
Megachilinae			
Anthidiini			
<i>Anthidium manicatum</i> (Linnaeus, 1758)	SO	A	GE
<i>Anthodioctes claudii</i> Urban, 1999	SO	A	GE
<i>Anthodioctes megachiloides</i> Holmberg, 1903	SO	A	GE
<i>Ctenanthidium gracile</i> Urban, 1993	SO	A	GE
<i>Dicranthidium</i> aff. <i>soniae</i> Urban, 1993	SO	A	GE
<i>Dicranthidium</i> sp. 01	SO	A	GE
<i>Epanthidium autumnale</i> (Schrottky, 1909)	SO	A	GE
<i>Epanthidium nectarinioides</i> (Schrottky, 1902)	SO	A	GE
<i>Hypanthidium divaricatum</i> (Smith, 1854)	SO	A	GE
<i>Moureanthidium paranaense</i> Urban, 1995	SO	A	GE
<i>Moureanthidium subarenarium</i> (Schwarz, 1933)	SO	A	GE
Megachilini			
<i>Coelioxys (Acrocoelioxys) toteca</i> Cresson, 1878	CL	NA	NA
<i>Coelioxys (Cyrtocoelioxys)</i> sp. 01	CL	NA	NA
<i>Coelioxys (Cyrtocoelioxys)</i> sp. 02	CL	NA	NA
<i>Coelioxys (Cyrtocoelioxys)</i> sp. 03	CL	NA	NA

Bee species	Behavior	Nesting site	Feeding habit
<i>Coelioxys (Glyptocoelioxys) cerasiopleura</i> Holmberg, 1903	CL	NA	NA
<i>Coelioxys (Glyptocoelioxys) chacoensis</i> Holmberg, 1903	CL	NA	NA
<i>Coelioxys (Glyptocoelioxys) sp. 01</i>	CL	NA	NA
<i>Coelioxys (Neocoelioxys) praetextata</i> Haliday, 1836	CL	NA	NA
<i>Coelioxys (Glyptocoelioxys) sp. 01</i>	CL	NA	NA
<i>Coelioxys (Neocoelioxys) praetextata</i> Haliday, 1836	CL	NA	NA
<i>Coelioxys (Neocoelioxys) sp. 01</i>	CL	NA	NA
<i>Megachile (Acentron) lentifera</i> Vachal, 1909	SO	A	GE
<i>Megachile (Acentron) sp. 01</i>	SO	A	GE
<i>Megachile (Austromegachile) facialis</i> Vachal, 1909	SO	A	GE
<i>Megachile (Austromegachile) susurrans</i> Haliday, 1836	SO	A	GE
<i>Megachile (Austromegachile) trigonaspis</i> Schrottky, 1913	SO	A	GE
<i>Megachile (Austromegachile) sp. 01</i>	SO	A	GE
<i>Megachile (Chrysosarus) cfr. tuberculifera</i> Schrottky, 1913	SO	A	GE
<i>Megachile (Chrysosarus) sp. 01</i>	SO	A	GE
<i>Megachile (Leptorachina) sp. 01</i>	SO	A	GE
<i>Megachile (Leptorachis) aetheria</i> Mitchell, 1930	SO	A	GE
<i>Megachile (Leptorachis) paulistana</i> Schrottky, 1902	SO	A	GE
<i>Megachile (Leptorachis) sp. 01</i>	SO	A	GE
<i>Megachile (Melanosarus) aff. brasiliensis</i> Dalla Torre, 1896	SO	A	GE
<i>Megachile (Moureapis) apicipennis</i> Schrottky, 1902	SO	A	GE
<i>Megachile (Moureapis) maculata</i> Smith, 1853	SO	A	GE
<i>Megachile (Moureapis) nigropilosa</i> Schrottky, 1902	SO	A	GE
<i>Megachile (Pseudocentron) curvipes</i> Smith, 1853	SO	A	GE
<i>Megachile (Tylomegachile) orba</i> Schrottky, 1913	SO	A	GE
<i>Megachile sp. 01</i>	SO	A	GE
<i>Megachile sp. 02</i>	SO	A	GE

Online Resource 6. Sample areas with their respective values of Shannon's diversity index (Diver), bee richness (Total), richness of each functional group: solitary behavior (Solit), eusocial (Eusoc), highly eussocial (High_eus), primitively eussocial (Prim_eus), nests in the ground (Ground), above ground (Above), Specialist (Specialist), Generalist (Generalist) and Cleptoparasite (Clepto). The areas are displayed from the lowest richness value to the highest.

Area s	Dive r	Tota l	Soli t	Eusoc	High_e us	Prim_eu s	Groun d	Above	Speciali st	Generali st	Clepto
1	3.34	62	35	26	7	19	27	34	4	57	1
2	3.75	111	70	36	11	25	57	49	15	91	5
3	3.80	115	74	32	6	26	61	45	15	91	9
4	4.10	119	77	36	11	25	67	46	18	95	6
5	4.13	121	79	33	9	24	60	52	18	94	9
6	4.11	123	86	30	8	22	64	52	18	98	7
7	3.96	125	84	33	7	26	69	48	22	95	8
8	3.67	130	89	35	9	26	76	48	25	99	6
9	3.97	131	88	32	9	23	74	46	25	95	11
10	4.21	134	94	31	5	26	83	42	28	97	9

Online Resource 7. Model selection for the analysis that presented the no effect model with $\Delta \text{AICc} < 2$, for bee responses with the of vegetation cover proportion among the predictors. We calculated the parameters of relative difference (ΔAICc), relative likelihood (w_i) and the Intercept of models. All these variables were selected at 750m (i.e., scale of effect) and we showed just the first three parameters.

Response variable/Predictor variable	ΔAICc	Weight (w_i)	Intercept (α)
Bee diversity			
No effect model	0.00	0.43	-
Isolation	1.21	0.23	-
Proportion of vegetation cover	1.41	0.21	-
Eusocial bee richness			
No effect model	0.00	0.83	-
Isolation	5.48	0.05	-
Proportion of arboreal cover	5.55	0.05	-
Primitively eusocial bee richness			
No effect model	0.00	0.95	-
Proportion of vegetation cover	7.46	0.02	-
Isolation	7.66	0.02	-
Highly eusocial bee richness			
No effect model	0.00	0.42	-
Edge density	1.07	0.24	-
Number of patches	2.08	0.15	-
Above ground nester bee richness			
No effect model	0.00	0.44	-
Isolation	0.86	0.29	-
Proportion of vegetation cover	1.21	0.24	-

Online Resource 8. Model selection for the analysis that presented the no effect model with $\Delta \text{AICc} < 2$, for bee responses with landscape diversity among the predictors. We calculated the parameters of relative difference (ΔAICc), relative likelihood (w_i) and the Intercept (α) of models. All these variables were selected at 750m (i.e., scale of effect) and we showed just the first three parameters.

Response variable/Predictor variable	ΔAICc	Weight (w_i)	Intercept (α)
Bee diversity			
Proportion of grassland	0.00	0.36	-
No effect model	0.53	0.28	-
Landscape Diversity	1.62	0.16	-
Eusocial bee richness			
No effect model	0.00	0.75	-
Edge density	4.89	0.06	-
Landscape Diversity	4.97	0.06	-
Primitively eusocial bee richness			
No effect model	0.00	0.93	-
Landscape Diversity	7.07	0.03	-
Proportion of grassland	7.73	0.02	-
Highly eusocial bee richness			
No effect model	0.00	0.45	-
Proportion of grassland	1.73	0.19	-
Isolation	2.26	0.14	-
Above ground nester bee richness			
No effect model	0.00	0.62	-
Landscape Diversity + Proportion of grassland	1.96	0.23	-
Landscape Diversity + Isolation	3.57	0.10	-

Online Resource 9. Proportion of vegetation cover (PLAND_VEG) and Shanon's diversity index of the landscape (SHDI) of each selected scale (km) for all sample areas, with the respective variance values of each scale.

Areas	PLAND_VEG_0.7 5	PLAND_VEG_ 1	PLAND_VEG_1. 5	PLAND_VEG_ 2	SHDI_0.7 5	SHDI_ 1	SHDI_1. 5	SHDI_ 2
1	7.02	4.96	3.72	4.47	0.32	0.24	0.19	0.22
2	45.73	51.25	51.00	47.22	1.09	0.93	0.71	0.61
3	38.68	27.84	15.73	10.81	1.04	1.01	0.92	0.70
4	53.83	33.95	21.98	17.52	1.11	1.12	1.06	1.03
5	57.45	19.76	11.91	10.28	1.29	1.25	1.17	1.14
6	40.59	33.94	30.89	20.34	1.04	0.83	0.57	0.44
7	33.47	56.56	41.64	35.41	0.91	0.66	0.46	0.41
8	76.93	52.29	48.12	47.13	1.27	1.24	1.20	1.09
9	72.67	66.34	49.90	43.08	1.10	1.08	1.10	1.14
10	71.07	65.40	63.14	54.34	1.35	1.27	1.12	1.01
Variance	456.48	414.88	393.49	336.84	0.08	0.10	0.12	0.12